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END SEMESTER EXAMINATION - 2025

Name of the Course: B.Tech.

Semester: V

Name of the Paper: Digital Signal Processing

Paper Code: **TEC 501**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
	Find the inverse Z- transform of the following using partial fraction method: $X(z) = \frac{1}{(1-1.5z^{-1}+0.5z^{-2})}$	
(a)	When a) ROC $ z > 1$ b) ROC $ z < 0.5$ c) ROC $0.5 < z < 1$	CO1
(b)	Determine the DTFT and Energy Density Spectrum of the sequence: $x(n) = \begin{cases} A; & 0 \leq n \leq L-1 \\ 0; & \text{Otherwise} \end{cases}$	
(c)	Illustrate the discrete time sinusoid properties with proof.	
Q2	(20 marks)	
(a)	State and prove the "Circular Time Shifting" property and "Parseval's Relation" of DFT.	CO2
(b)	Proof that the multiplication of two DFTs is equivalent to circular convolution.	
(c)	Compute N = 8 Point DFT of given below sequence using DITFFT algorithm: $x(n) = \{0, 1, 2, 3, 4, 5, 6, 7\}$	
Q3	(20 marks)	
(a)	Prove the Direct Form II structure of the IIR digital filter. Draw its Signal flow graph and its Transposed Structure.	CO3
(b)	Determine the cascade and parallel realization of the following systems: $H(z) = \frac{3(4+5z+2z^2)}{(2z+1)(z+2)}$	
(c)	Prove and draw the structure for linear phase FIR Filter for M even. Determine and draw the linear phase realization of FIR filter, whose system function is: $H(z) = \left(\frac{5}{8} + z^{-1} + \frac{5}{8}z^{-2}\right)$	
Q4	(20 marks)	
(a)	Prove the 'Bilinear Transformation Technique' for design of IIR Filter.	CO4
(b)	Prove the "Impulse Invariant" transformation technique for IIR digital filter. Convert the analog bandpass filter with system function $H(s) = \frac{1}{(s + 0.1)^2 + 9}$ into a digital IIR filter by use of the backward difference for the derivatives.	

(c)	Explain the design steps of "Butterworth filter".	
Q5	(20 marks)	
(a)	Explain Hamming, Hanning and Blackman, Rectangle, Bartlett and Kaiser windows for design of FIR filter.	CO5
(b)	The desired response of low- pass filter is $H_d(e^{j\omega}) = e^{-j2\omega}; -\pi/4 < \omega < \pi/4$ $= 0; \pi/4 < \omega \leq \pi$ Determine $H(e^{j\omega})$ if window function is defined as $w(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$	
(c)	What is "Gibbs phenomenon"? Why this phenomenon is developed in rectangular window and it is eliminated or reduced? Discuss the windowing method of FIR filter design.	